27,求序列的z变换

$$x(n) = a^n cos(\pi n)u(n)$$
$$x(n) = [2^{n-1} - (-2)^{n-1}]u(n)$$

```
syms n z a;
x1=(-a)^n*heaviside(n);
x2=(2^(n-1)-(-2)^(n-1))*heaviside(n);
X1=ztrans(x1,n,z);
X2=ztrans(x2,n,z);
pretty(X1);
pretty(X2);
```

得到输出结果为，依次为两问的z变换值

```
1      a
- - -----
2    a + z

  1       1      1
----- - ----- + -
z - 2   z + 2   2
```

28，求序列反变换

$$X(z) = \frac{8z - 19}{z^2 - 5z + 6}$$
$$X(z) = \frac{z(2z^2 - 11z + 12)}{(z-1)(z-2)^2}$$
$$X(z) = \frac{1 - 2z^{-1}}{z^{-1} - 2}$$

```
syms n z;
X1=(8*z-19)/(z^2-5*z+6);
X2=z*(2*z^2-11*z+12)/((z-1)*(z-2)^2);
X3=(1-2/z)/(1/z-2);
x1=iztrans(X1,z,n);
x2=iztrans(X2,z,n);
x3=iztrans(X3,z,n);
pretty(x1);
pretty(x2);
pretty(x3);
```

得到输出结果为，依次为三问的z反变换值

$$\frac{3\,2^n}{2} + \frac{5\,3^n}{3} - \frac{19\,\mathrm{kroneckerDelta}(n,\ 0)}{6}$$

$$3 - 2^n\,(n - 1) - 2\,2^n$$

$$\frac{3\left(\frac{1}{2}\right)^n}{2} - 2\,\mathrm{kroneckerDelta}(n,\ 0)$$